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4 March 1977
SUPERSEDING
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MILITARY SPECIFICATION

WATCH, WRIST: SUBMERSIBLE AND NAVIGATION

1. SCOPE

1.1 This specification contains the requirements for a wrist watch with an elapsed time ring capable of operating at a depth of 1200 feet and at an altitude of 35,000 feet which is not affected by operation in a magnetic field of 125 gauss. A strap is furnished with the watch.

1.2 Classification.- Wrist watches covered by this specification shall be of the following types and classes:

Type I - Symboled dial (See Figure 3)

Style N - Navigation Elapsed Time Ring (See Figure 2)
Style S - Submersible Elapsed Time Ring (See Figure 8)

Type II - Numbered dial (See Figure 4)

Class A - With luminous dial and hands
Class B - Without luminous dial and hands

NOTE: Styles are applicable to Type I only.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-T-360 Time Measuring Instruments; Packaging of

Military

MIL-S-46383 Strap, Wrist: Instrument

FSC 6645

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in approving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., PA 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electric Component Parts

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS
US Nuclear Regulatory Commission, Rules and Regulations
Title 10 Chapter I, Parts 30 and 32

(Application for copies should be addressed to Superintendent of Documents, US Government Printing Office, Washington, D. C. 20402)

3. REQUIREMENTS

3.1 First article.- This specification provides for first article inspection as specified herein.

3.2 Design.- The watch shall be designed to operate at a water depth of 1200 feet, at an altitude of 35,000 feet and in a magnetic field of 125 Gauss. A self-winding, crown-set movement, which can be manually wound, containing a second hand stop mechanism, a manually locking crown, a minimum of 17 jewels and a shock absorbing device shall drive concentrically mounted hour, minute and sweep-second hands. The watch shall have an outer elapsed time ring (bezel) and include a woven nylon strap.

3.3 Materials.- All materials shall be of uniform quality and free of defects which might impair the functioning or accuracy of the watch. Material which is not specified by a definite material specification shall be of a composition and quality that will enable the watches to meet all requirements of this specification.

3.3.1 Metal parts.- All external metal parts shall be corrosion-resistant steel. Internal metal parts which are susceptible to corrosion

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shall be protected by a lubricant except those parts whose proper functioning would be detrimentally affected.

3.3.2 Paint.

3.3.2.1 White paint.- All areas of the dials, hands, and elapsed time ring for which White, Color No. 37875 (see Figures 2, 3, 4, 5 and 8) is specified, shall be a paint or lacquer containing a minimum of 25% pigment. The pigment shall contain a minimum of 90% Titanium Dioxide (TiO_2).

3.3.2.2 Self-luminous paint.- The self-luminous paint for dials, hands and elapsed time ring (bezel) shall consist of a phosphorescent material, ceramic-bound spheres of Promethium ($Pm147$) and a binder.

NOTE: This requirement is applicable only to Class A watches.

3.3.2.2.1 Types I and II (Class A).- The quantity of Promethium 147 in the self-luminous paint shall not exceed 100 microcuries per watch as specified for timepieces in Title 10, Chapter I, Part 30 of the Code of Federal Regulations (see 6.3 and Figures 2, 3, 4, 5 and 8).

3.3.2.2.2 Undercoat.- A paint or lacquer undercoat shall be applied to the total areas of the dial, hands and elapsed time ring containing self-luminous paint (see Figures 2, 3, 4, 5 and 8). The undercoat shall be a minimum thickness of 0.001 inch and contain a minimum of 25% pigment. The pigment shall contain a minimum of 90% Titanium Dioxide (TiO_2).

3.4 Movement.- The movement shall be crown-wound and self-winding, crown set, with the crown located at the three o'clock position of the dial.

3.4.1 Self-winding rotor.- The self-winding rotor shall operate through 360°, in both clockwise and counterclockwise directions.

3.4.2 Mainspring.- The material for the mainspring shall be a corrosion-resistant "nonbreakable" alloy.

3.4.3 Running time.- When fully wound, the mainspring shall drive the movement a minimum of 36 hours without being wound or the self-winding mechanism being activated.

3.4.4 Hairspring and balance wheel assembly.- The movement shall have a temperature compensated hairspring and a solid monometallic, non-magnetic balance wheel.

3.4.5 Second hand stop mechanism.- Pulling the crown from the winding position to the setting position shall result in stopping of the movement and hands. Rotation of the crown, in the setting position, shall allow the minute and hours to be set without any movement of the second hand. The depressing of the crown to the winding position shall result in the movement and hands resuming the measurement of time.

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3.5 Case.- The case shall be in accordance with Figure 1 and designed to limit access to the movement except to authorized repair personnel.

3.5.1 Case bars.- The bars of the case through which the strap is inserted shall be an integral part of the case or solid bars permanently secured to the case and capable of withstanding a pull of 40 pounds, ± 1/2 pound.

3.5.2 Crown.- The crown shall have a straight knurl, conform to the dimensions in Figure 1, and include a locking mechanism. The release of the locking mechanism, by rotating counterclockwise, shall result in the crown moving into the winding position.

3.5.3 Finish.- The case, case bars and crown shall have a dull non-reflecting finish.

3.5.4 Crystal.- The crystal shall be clear, uncolored and free from bubbles or other imperfections which may interfere with reading of the watch, after being subjected to a depth of 1200 feet and an altitude of 35,000 feet. The crystal shall be recessed within the elapsed time ring.

3.5.5 Elapsed time ring.- Unless otherwise specified, the elapsed time ring shall conform to Style N (Figure 2). The ring shall have an outer diameter 0.025 to 0.030 inches less than the outer diameter of the watch case. The ring shall permit rotation and setting by pressure of the fingers and be designed to assure that the ring does not move when exposed to shock and vibration (see 3.9.7 and 3.9.11). The outer edge shall be serrated (see Figures 2 and 8).

3.5.5.1 Finish.- The elapsed time ring shall have a dull non-reflecting finish.

3.5.5.2 Style N ring.- The Style N ring, graduated in hour and minutes, shall conform to Figure 2.

3.5.5.3 Style S ring.- The Style S ring, graduated in minutes and seconds shall conform to Figure 8.

3.6 Dial.- The dial for the Type I watch shall conform to Figure 3. The dial for the Type II watch shall conform to Figure 4. The dial face, with markings, shall have a durable transparent protective coating.

3.7 Hands.- There shall be three hands, one each to indicate the hour, minute and second. The hands shall conform to Figure 5.

3.8 Strap.- The strap shall be a Type III as specified in MIL-S-46383. The color of the strap shall be black.

3.9 Performance.

3.9.1 Synchronization.- The hands shall be synchronized to eliminate the possibility of error in reading correct time. The hour hand shall in-

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dicate the correct time within the tolerance indicated for each dial, when the minute hand is at any hour, from 1 through 12:

a. Types I: ± six (6) degrees
b. Types II: ± one (1) dial graduation

3.9.2 Crown.- The locking mechanism shall remain in the locked position when subjected to shock (see 3.9.11), vibration (see 3.9.7), a pull of 10 pounds, ± 0.25 pound and a torque of 8 inch-ounces, ± 1/2 inch-ounce.

3.9.2.1 Setting.- The crown shall be capable, while being moved from winding to set position, of withstanding a pull of seven (7) pounds, ± 0.25 pound.

3.9.3 Elapsed time ring.

3.9.3.1 Design.- The elapsed time ring shall be designed to assure that the outer edge (shoulder) of the crystal is recessed within the elapsed time ring.

3.9.3.2 Torque.- The elapsed time ring shall move only when a torque of 26 ± 6 inch ounces is applied.

3.9.3.3 Displacement.- The elapsed time ring shall be capable of withstanding, without damage, two (2) forces, each of ten (10), ± 1/2 pounds, applied to the lower side of the elapsed time ring. The forces shall be applied alternately to opposite sides of the elapsed time ring with the index (triangle) of the elapsed time ring at the 3 o'clock position of the dial.

3.9.4 Isochronism.- In the dial up position at 23°C, ± 10°C, the difference in rate (see 6.2.2) in a 24 hour period, between the first four hours and the last four hours, shall not exceed five (5) seconds.

3.9.5 Magnetism.- While running, the daily rate shall not exceed +15 seconds, -5 seconds after being subjected to a magnetic field of 125 Gauss, ± 1.0 Gauss, with the crown of the watch parallel to the direction of the magnetic field.

3.9.6 Brightness (Class A only).- The watch shall be capable of being distinctly read in darkness at a minimum distance of twelve (12) inches from the eyes of a dark adapted observer having normal vision.

3.9.7 Vibration.- While running, the watch shall not be damaged by ninety (90) minutes of simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion), the frequency being varied uniformly between the approximate limits of 10 and 55 hertz (Hz). The entire frequency range, between 10 and 55 Hz and return to 10 Hz, shall be traversed in approximately one (1) minute. The motion shall be applied for thirty (30) minutes in each of the following directions:

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a. Perpendicular to dial
b. Through the 6 and 12 positions of the dial
c. Through the 3 and 9 positions of the dial

3.9.8 Storage.- The watch shall show no evidence of damage affecting serviceability after being subjected to storage temperatures of -53°C and +65°C for 24 hours at each temperature.

3.9.9 Altitude.- Watches shall show no evidence of damage when subjected to an altitude of 35,000 feet for a minimum of sixty (60) minutes.

3.9.10 Accuracy.- After meeting the requirements of 3.9.1 to 3.9.9 inclusive, the watches shall have a daily rate (see 6.2.3) and mean daily rate (see 6.2.4) as specified in 3.9.10.1 and 3.9.10.2. The daily and mean daily rates shall be determined in the dial-up and crown-down positions.

3.9.10.1 Daily rate.- The daily rate, in each of the specified positions, at 23°C, ± 10°C, shall not exceed +15 seconds, -5 seconds; at 4°C ± 1°C, shall not exceed +21 seconds, -11 seconds and at 52°C, ± 1°C, shall not exceed +28 seconds, -18 seconds.

3.9.10.2 Mean daily rate.- The mean daily rate, in each of the specified positions, at 23°C, ± 10°C, shall not exceed +15 seconds, -5 seconds.

3.9.11 Shock.- The watch shall show no evidence of damage after one uncontrolled drop, while running, from a height of two (2) feet, 6 inches, onto a block of hard wood. The daily rate, in each of the positions specified in 3.9.10 shall not exceed, at 23°C, ± 10°C, +20 seconds, - 10 seconds.

3.9.12 Waterproofness.- Watches shall show no evidence of damage and reject the entry of water at a depth of 1200 feet for a minimum of sixty (60) minutes. The daily rate, in each of the positions specified in 3.9.10 shall not exceed, at 23°C, ± 10°C, +20 seconds, -10 seconds.

3.10 Identification.- The back of each case shall be permanently marked as specified in Figure 6. The date (month and year) to be included in the identification shall be the date of acceptance by the Government. The marking for the month shall be the first three (3) letters of the month and the marking for the year shall be in full, e.g., December 1976 is DEC 1976. The serial numbers will be assigned by the contracting officer (see 6.1). Serial numbers of rejected watches shall not be reused.

3.11 Operating instructions.- An operating instruction, conforming to Figure 7, shall be furnished with each watch. The instruction shall be printed on 8 x 10-1/2, 20 pound white sulphite paper.

3.12 Workmanship.- Workmanship shall be of a quality consistent with the highest instrument production standards and practices., All finished surfaces shall be protected against corrosion or damage during manufacture prior to delivery. All surfaces shall be free of burrs and sharp edges. All material shall be sound, of uniform quality and condition, and free

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from seams, cracks and other defects which may adversely affect the strength, endurance or wear resistance of the watch. Any material which has been treated in any manner to conceal defects therein, shall not be offered for Government acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Quality assurance terms and definitions shall be as listed in MIL-STD-109.

4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection.- First article inspection shall be as specified in the contract or purchase order. First article inspection shall be in conformance with the first article tests listed in Table I.

4.3.1 Samples.- Four (4) watches shall be subjected to first article inspection. The watches shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production.

4.3.2 Failure.- Failure of any sample watch to pass all the tests listed in Table I is cause for rejection of all samples submitted at one time for first article inspection.

TABLE I - FIRST ARTICLE INSPECTION

Examination or Test	Requirement Paragraph	Method Paragraph
Design	3.2	4.8.3
Materials	3.3	4.8.1
Metal parts	3.3.1	4.6.2
Luminous material (Note 1)	3.3.2.2, 3.3.2.2.1	4.8.2.1
Undercoat (Note 1)	3.3.2.2.2	4.8.2.2
Movement	3.4	4.8.3
Self-winding rotor	3.4.1	4.6.2
Running time	3.4.3	4.8.4.1, 4.8.4.2

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TABLE I - FIRST ARTICLE INSPECTION (Cont'd)

Examination or Test	Requirement Paragraph	Method Paragraph
Hairspring and balance wheel assembly	3.4.4	4.8.5
Second hand stop mechanism	3.4.5	4.9.1.3.2
Case	3.5	4.8.6
Case bars	3.5.1	4.8.7
Crown	3.5.2	4.8.8
Finish	3.5.3, 3.5.5.1	4.8.9
Crystal	3.5.4	4.8.10
Elapsed time ring		
Style N	3.5.5.2	4.8.11.1
Style S	3.5.5.3	4.8.11.2
Dial	3.6	4.8.12
Hands	3.7	4.8.13
Strap	3.8	4.8.14
Synchronization	3.9.1	4.9.1.3.1
Crown	3.9.2, 3.9.2.1	4.9.1.1, 4.9.1.2
Elapsed time ring	3.9.3.1	4.9.2
Torque	3.9.3.2	4.9.2.1
Displacement	3.9.3.3	4.9.2.2
Isochronism	3.9.4	4.9.3
Magnetism	3.9.5	4.9.4
Brightness (Note 1)	3.9.6	4.9.5
Vibration	3.9.7	4.9.6
Storage	3.9.8	4.9.7
Altitude	3.9.9	4.9.8
Accuracy	3.9.10, 3.9.10.1, 3.9.10.2	4.9.10
Shock	3.9.11	4.9.11
Waterproofness	3.9.12	4.9.12
Identification	3.10	4.9.13
Operating instructions	3.11	4.9.14
Workmanship	3.12	4.9.15

NOTE 1: Applicable only to Class A watches.

4.4 Quality conformance inspection.- Quality conformance inspection shall be in accordance with the examinations and tests listed in Tables II and III.

4.4.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation and presentation of lots shall be in accordance with "Submission of Product" and "Drawing of Samples" specified in MIL-STD-105.

4.4.2 Classification of examinations and tests.- Examinations and tests shall be performed on a defect (individual characteristic) basis in

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accordance with MIL-STD-105 and the Inspection Level and Sampling Plans specified in Tables II and III. Examination and tests for packaging, packing and marking shall be in accordance with PPP-T-360 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE II - CLASSIFICATION OF DEFECTS

Level II of Table I with Sampling Plan Table II-A of MIL-STD-105

CRITICAL: AQL 0.0	Requirement	Method
1. Waterproofness	3.9.12	4.9.12
2. Torque	3.9.3.2	4.9.2.1
MAJOR: AQL 1.5		
101. Second hand stop mechanism	3.4.5	4.9.1.3.2
102. Case bars	3.5.1	4.8.7
103. Crystal	3.5.4	4.8.10
104. Elapsed time ring		
Style S	3.5.5.2	4.8.11.1
Style N	3.5.5.3	4.8.11.2
105. Synchronization	3.9.1	4.9.1.3.1
106. Displacement	3.9.3.3	4.9.2.2
107. Storage	3.9.8	4.9.7
MINOR: AQL 2.5		
201. Strap	3.8	4.8.14
202. Identification	3.10	4.9.13
203. Operating instructions	3.11	4.9.14
204. Workmanship	3.12	4.9.15

TABLE III - CLASSIFICATION OF DEFECTS

Level S-4 of Table I with Sampling Plan Table II-A of MIL-STD-105

CRITICAL: None		
MAJOR: AQL 4.0	Requirement	Method
109. Isochronism	3.9.4	4.9.3
110. Magnetism	3.9.5	4.9.4
111. Brightness (Note 1)	3.9.6	4.9.5
112. Vibration	3.9.7	4.9.6

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TABLE III - CLASSIFICATION OF DEFECTS (Cont'd)

		Requirement	Method
113.	Altitude	3.9.9	4.9.8
114.	Accuracy	3.9.10.1	4.9.9
115.	Shock	3.9.11	4.9.11

NOTE 1: Applicable only to Class A watches.

4.5 Test equipment.- Accuracy of the watch shall be determined by a mechanical, electric or electronic time measuring instrument having a daily rate of ± two (2) seconds per day.

4.6 Visual examination.- The watch shall be examined visually for conformance to workmanship and requirements as specified in Tables II and III.

4.6.1 Mechanical examination.- The watch shall be examined for conformance to dimensional requirements and control operations.

4.6.2 Certification.- Certification shall state that the watch is in full compliance with the specified requirements and that workmanship and materials conform with recognized standards of commercial quality. The certification shall reference the contract or purchase order and shall be signed by a responsible officer of the contractor. The certification shall contain the following statement: "As the authorized representative of the contractor, the undersigned warrants and represents that all the information supplied herewith is true and accurate".

4.7 Test conditions.

4.7.1 Atmospheric.- Unless otherwise specified, the tests shall be performed at 23°C ± 10°C, at barometric pressure of 28 to 31 inches of mercury and a maximum relative humidity of 80 percent and without the strap.

4.7.2 Operation.- Unless otherwise specified, watches shall be operating (running) during and following all tests.

4.7.3 Accuracy.- Unless otherwise specified, watches shall be accurate within +15 seconds, - 5 seconds, during and following the tests of 4.9.

4.7.4 Crown.- Unless otherwise specified, the crown shall be in the locked position during and following the tests of 4.9.

4.8 Test methods.

4.8.1 Materials.- Acceptance of materials shall be by certification. Where defects or inferior quality is evident, and the Government deems a

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material analysis necessary, the contractor will be requested to submit samples to the contracting officer for analysis and approval.

4.8.2 Luminous material

4.8.2.1 Class A.- The luminous material for Class A watches containing Promethium 147 shall be tested as specified for timepieces in Part 32 of Title 10, Chapter I of the Code of Federal Regulations. Certification of compliance with the specific license (see 6.3) shall be provided for each batch of material processed for application to the dial, hands and elapsed time ring.

4.8.2.2 Undercoat.- The dials, hands and elapsed time ring of Class A watches shall be certified (see 4.6.2) to have an undercoat with a minimum thickness of 0.001 inch containing a minimum of 25% pigment that contains a minimum of 90% Titanium Dioxide (TiO_2).

4.8.3 Movement (design).- The watches shall be visually examined to determine compliance with design, including movement (self-winding with manual wind capabilities).

4.8.4 Running time.

4.8.4.1 Manually wound.- The watches shall be fully wound, manually, and then shall run continuously a minimum of 36 hours, without rewinding.

4.8.4.2 Self-wound.- The watches shall be fully wound by the self-winding mechanism, and then shall run continuously a minimum of 36 hours, without additional motion of the self-winding rotor.

4.8.5 Hairspring and balance wheel assembly.- The hairspring and balance wheel unit shall be considered acceptable if the watch is capable of passing the test of 4.8.4 and 4.9.4.

4.8.6 Case.- The case shall be inspected visually and dimensionally to determine conformance to Figure 1.

4.8.7 Case bars.- A visual examination shall be made to insure that the bars are an integral part of the case. The strap (see 3.8) shall be looped successively around each case bar. With the watch in a secured position and the strap held in a position that will not exert pressure on the buckle or the keeper of the strap, a pulling force of 40 pounds, $\pm$ 1/2 pound, shall be applied to the case bar via the strap without the case bar permanently bending or causing other damage to the bar or case.

4.8.8 Crown.- The crown shall be inspected visually and dimensionally to determine conformance to Figure 1. The locking mechanism shall be checked to assure that it is activated when the crown is pressed toward the case and turned clockwise. The crown shall also be checked to assure that turning the crown counterclockwise, while it is in the locked position, shall result

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in the release of the locking mechanism and the moving of the crown into the winding position.

4.8.9 Finish.- The case, case bars, crown and elapsed time ring shall be visually examined for a dull, nonreflecting finish.

4.8.10 Crystal.- The crystal, after the tests of 4.9.7, 4.9.8 and 4.9.11 shall be visually examined for any imperfections which would interfere with reading of the watch, and shall show no evidence of scratches or digs.

4.8.11 Elapsed time ring.

4.8.11.1 Style S. The Style S ring shall be inspected visually and dimensionally to determine conformance with Figure 8 and 3.9.3.1.

4.8.11.2 Style N.- The Style N ring shall be inspected visually and dimensionally to determine conformance with Figure 2 and 3.9.3.1.

4.8.12 Dial.- The dial shall be visually and dimensionally examined to determine conformance to Figure 3 for the Type I dial and to Figure 4 for the Type II dial.

4.8.13 Hands.- The hands shall be visually and dimensionally examined to determine conformance to Figure 5.

4.8.14 Strap. The strap shall be certified to be in conformance with Type III of MIL-S-46383.

4.9 Performance.

4.9.1 Crown.

4.9.1.1 Locked position.- In the locked position the functions of the crown shall not be impaired when subjected to:

a. A torque of eight (8) inch-ounces, ± 1/2 inch-ounce.

b. A direct force (pull) of ten (10) pounds, ± 0.25 pound. The application of the direct force, in addition to not resulting in damage, shall not result in the crown moving into the winding or setting position.

4.9.1.2 Setting.- The crown shall not be damaged by a direct force (pull) of 7 pounds, + _ 0.25 pound, applied while moving the crown from winding to setting position.

4.9.1.3 Setting position.

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4.9.1.3.1 Synchronization.- In the setting position the crown shall be turned and readings observed with the minute hand at 12. At any hour the readings shall indicate the correct time within ± 6 degrees (one dial graduation).

4.9.1.3.2 Second hand stop mechanism.- In the setting position the crown shall be turned to move the hour and minute hands forward two hours. This action shall be repeated five times. The second hand moving with the crown in the setting position shall be cause for rejection.

4.9.2 Elapsed time ring.- The elapsed time ring shall be visually examined to assure that the outer edge (shoulder) of the crystal is recessed within the elapsed time ring.

4.9.2.1 Torque.- The elapsed time ring shall move only when subjected to a torque applied clockwise and counterclockwide, of 26 ± 6 inch ounces.

4.9.2.2 Displacement.- With the index (triangle) at the 3 o'clock position of the dial, a force of ten (10) pounds, ± 1/2 pound, shall be applied successively, 180° apart, to the lower side of the elapsed time ring without the ring being damaged or separating from the case.

4.9.3 Isochronism.- In the dial-up position, the watch shall be fully wound and operate for four hours. The watch shall again be fully wound and the rate recorded. Without rewinding, the rate shall be recorded at 4 hours, 20 hours and 24 hours. The difference in rate (see 6.2.2) between the first four hours and the last four hours shall not exceed five seconds.

4.9.4 Magnetism.- The watch shall be subjected, for ten minutes, to a magnetic field of 125 Gauss, ± 1.00 Gauss, with the crown of the watch parallel to the direction of the magnetic field.

4.9.5 Brightness (Class A only).- Brightness shall be tested in a totally darkened area by an observer who has been dark adapted. The observer shall be able to read the time, with the watch a minimum of twelve (12) inches from the eyes of the observer. The watch tested shall have been kept in total darkness a minimum of 8 hours prior to this test.

4.9.6 While running, the watch shall be subjected to Method 201 of MIL-STD-202 for 30 minutes in each of the following directions:

a. Perpendicular to dial
b. Through the 6 and 12 positions of the dial
c. Through the 3 and 9 positions of the dial

4.9.7 Storage.- The watch shall not be damaged when subjected to:

a. -53°C (± 1°C) for 24 hours
b. +65°C (± 1°C) for 24 hours

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NOTE: Watches shall not be run during the storage tests. The watches are exempt from being tested for accuracy at the completion of this test.

4.9.8 Altitude.- The watch shall be subjected to air pressure equivalent to that experienced at 35,000 feet (550mm Hg Vacuum) for 60 minutes without damage.

4.9.9 Accuracy.- After completion of all tests, except shock (4.9.11) and waterproofness (4.9.12), the daily rate (see 6.2.3) of the watch, at 23°C (± 10°C), in the positions of (1) dial up and (2) crown down, shall not exceed +15 seconds, -5 seconds.

4.9.10 Accuracy (first article only).- After completion of all tests, except shock (4.9.11) and waterproofness (4.9.12), the mean daily rate (see 6.2.4) in the positions of (1) dial up and (2) crown down at 23°C, ± 10°C, shall not exceed +15 seconds, -5 seconds. In the same positions the daily rate at 4°C ± 1°C, shall not exceed +21 seconds, -11 seconds and at 52°C, ± 1°C, shall not exceed +28 seconds, -18 seconds. The watches shall be subjected to each test temperature for at least four hours prior to the test. The watches shall be wound at the beginning of each test and then once every 24 hours during the test.

4.9.11 Shock.- While running, the watch shall be dropped, uncontrolled, from a height of two feet, six inches onto the end grain of beech, oak or hard maple, the size of which shall be a minimum of four inches. square by four inches thick. The watch, at the conclusion of the test, shall show no evidence of damage. The daily rate (see 6.2.3), after the test, at 23°C, ± 10°C, in the positions of (1) dial up and (2) crown down shall not exceed +20 seconds, -10 seconds.

4.9.12 Waterproofness.- The watch shall be immersed in distilled water. The water shall be subjected to a pressure of 521 pounds per square inch for 60 minutes. After removal from water, the watch shall be subjected to a temperature of +45°C, ± 1°C, for thirty minutes. After the test the watch shall not show evidence of damage or condensation within the case. The daily rate (see 6.2.3), after the test, at 23°C, ± 10°C, in the positions of (1) dial up and (2) crown down, shall not exceed +20 seconds, -10 seconds. The waterproofness test shall be the last test to which the watches are subjected.

4.9.13 Identification.- All numbers, names and location of identification marking shall be inspected for compliance with Figure 6.

4.9.14 Operating instructions.- The operating instructions shall be inspected for compliance with Figure 7.

4.9.15 Workmanship.- Quality of workmanship in conjunction with industry standard practices shall be inspected, at the discretion of the Government, during in-process and on completed watches to insure that watches are continually produced in accordance with 3.12.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- Packaging and packing shall be as specified for Packaging Group I of PPP-T-360. The level of protection shall be as specified in the purchase order or contract. The operating instructions (see 3.11) shall be included in each unit package.

NOTE: The triangle of the elapsed time ring shall be at 12 of the dial and the crown locked when placed in the unit package.

5.2 Marking.- In addition to the marking requirements specified in PPP-T-360, the unit package, intermediate package and shipping container shall be marked with the date of acceptance by the Government.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type, style and class of watch required.
   (1) For Type I watches, the class must be specified (see 3.5.5.2 (Figure 8) and 3.5.5.3 (Figure 2).
c. Selection of applicable levels of packaging and packing.
d. Applicable stock number.
e. List of serial numbers to be assigned (see 3.10).

6.2 Definitions of terms used.

6.2.1 Accuracy error notation.- Where algebraic signs are used to denote the direction of timekeeping accuracy error, the plus (+) sign represents "fast" and the minus (-) sign "slow".

6.2.2 Rate.- Rate is the error, in seconds, between any two readings of the watch.

6.2.3 Daily rate.- The term "daily rate" is used synonymously with the terms "daily error" and "daily accuracy".

6.2.4 Mean daily rate.- Mean daily rate is the arithmetic average of individual daily rates (daily errors) with proper regard to algebraic signs in the summation. Unless otherwise specified, the mean daily rate shall be for three (3) consecutive days operation.

6.3 Luminous material.- Prospective bidders shall be cognizant of the need for a license in the application of Promethium 147 as required in Sections 30.15 and 32.14 of CFR, Title 10, as referenced in 2.2.

Custodians
Army - MU
Navy - SH
Air Force - 99

Preparing activity:
Army - MU

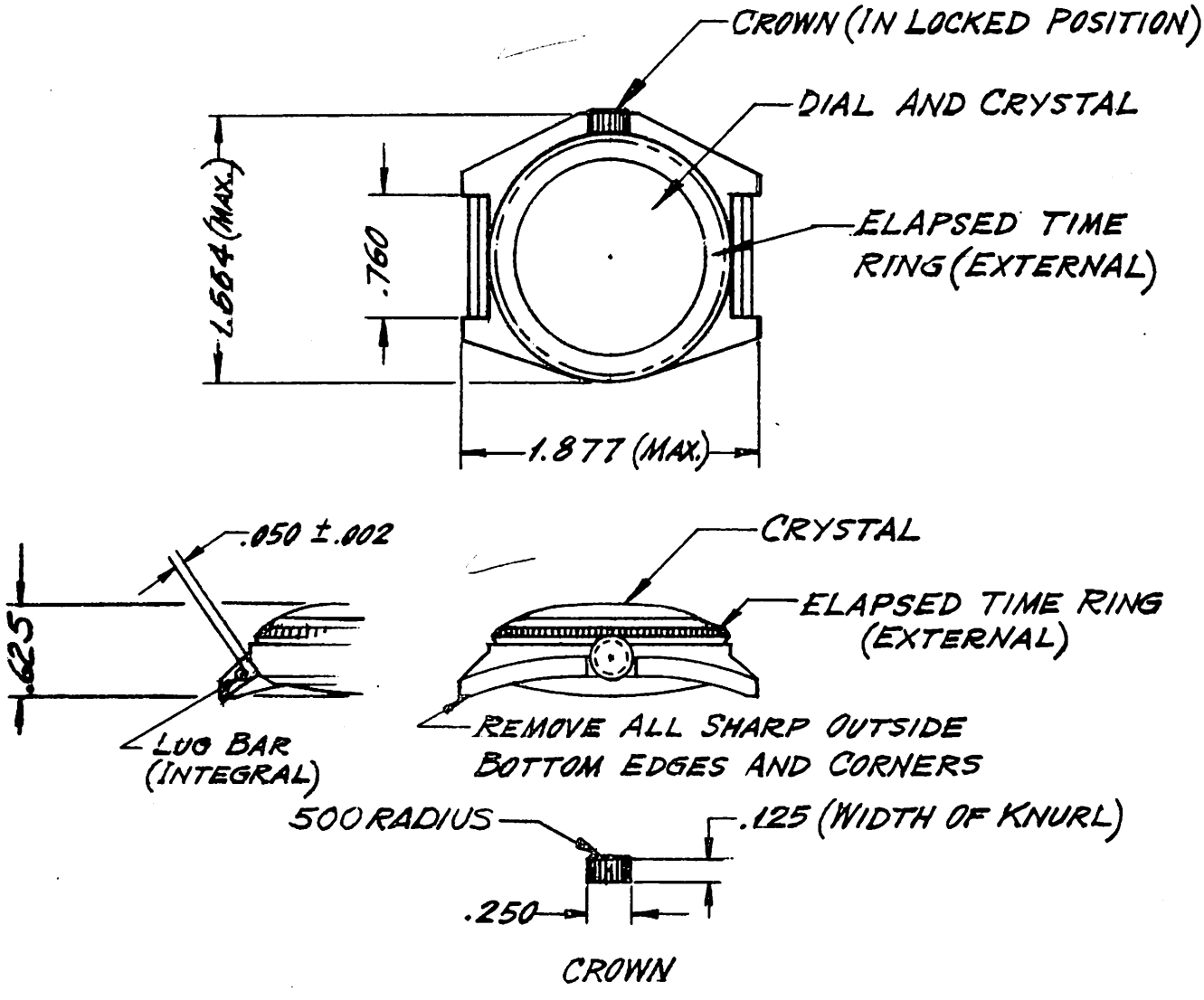
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Review:
Navy - OS, AS
DSA - GS

User:
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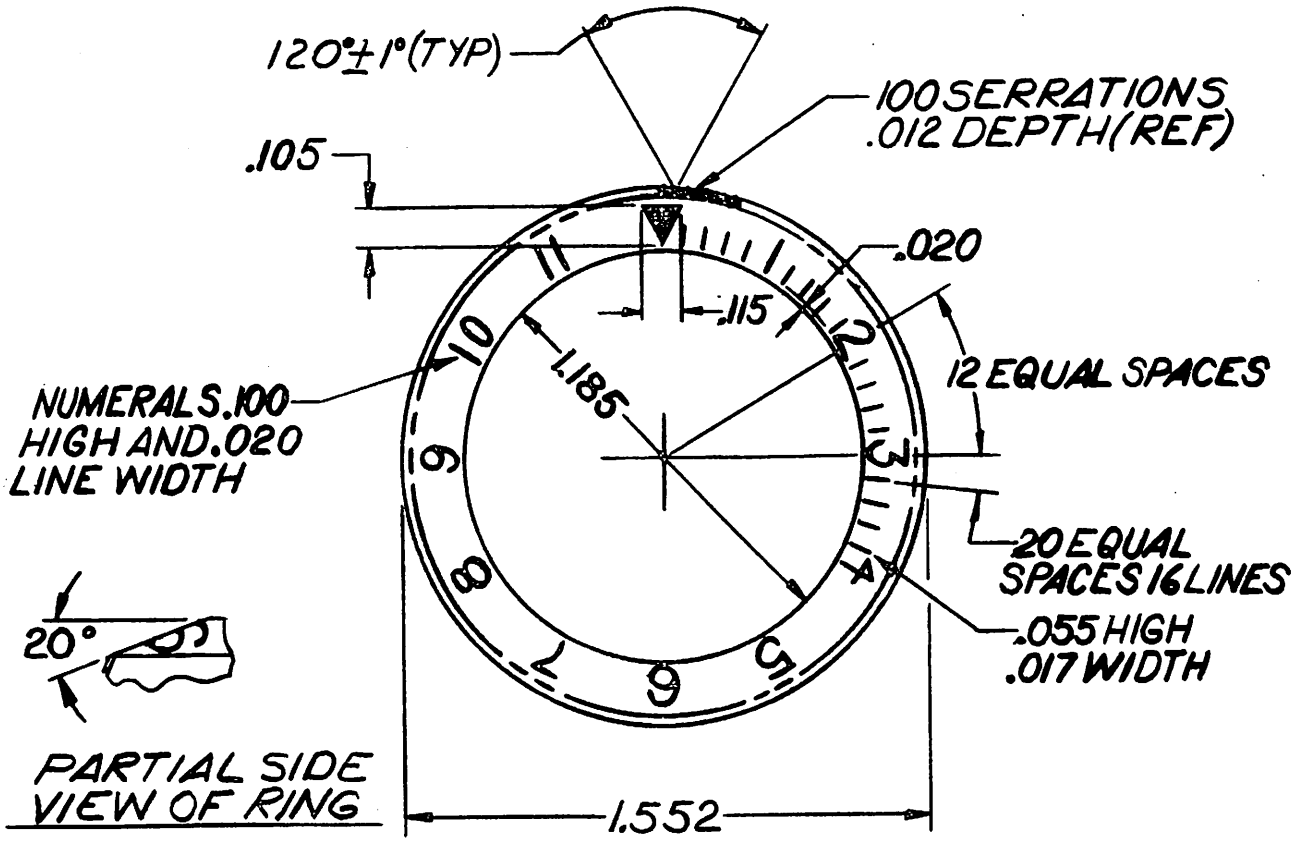
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DIMENSIONS IN INCHES
TOLERANCES ± .010 (UNLESS OTHERWISE SPECIFIED)

FIGURE 1

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NOTES:

1. BACKGROUND IN ACCORDANCE WITH FED-STD-595, COLOR BLACK NO.37038.

2. NUMBERS AND GRADUATIONS IN ACCORDANCE WITH FED-STD-595, COLOR WHITE NO.37875.

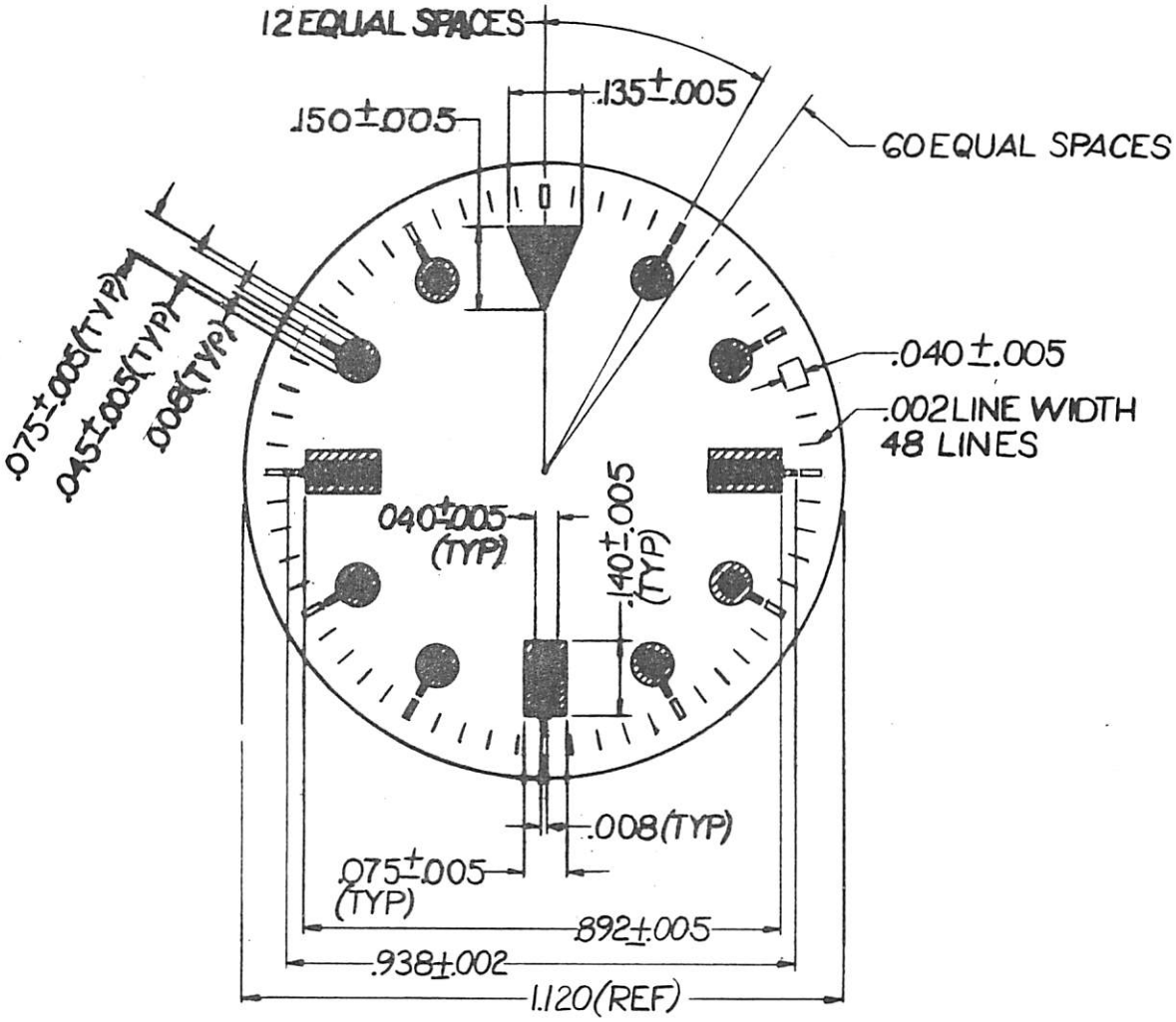
3. SHADED TRIANGLE:
   CLASS A: LUMINOUS GREEN (SEE 3.3.2, 3.3.2.1 AND 3.3.2.2)
   CLASS B: FED-STD-595, COLOR WHITE NO.37875

4. MINUTE GRADUATIONS BETWEEN TRIANGLE AND 4 ONLY

DIMENSIONS IN INCHES UNLESS
OTHERWISE SPECIFIED
TOLERANCES ±.010
ANGLES ± 1°

FIGURE 2

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TYPE I

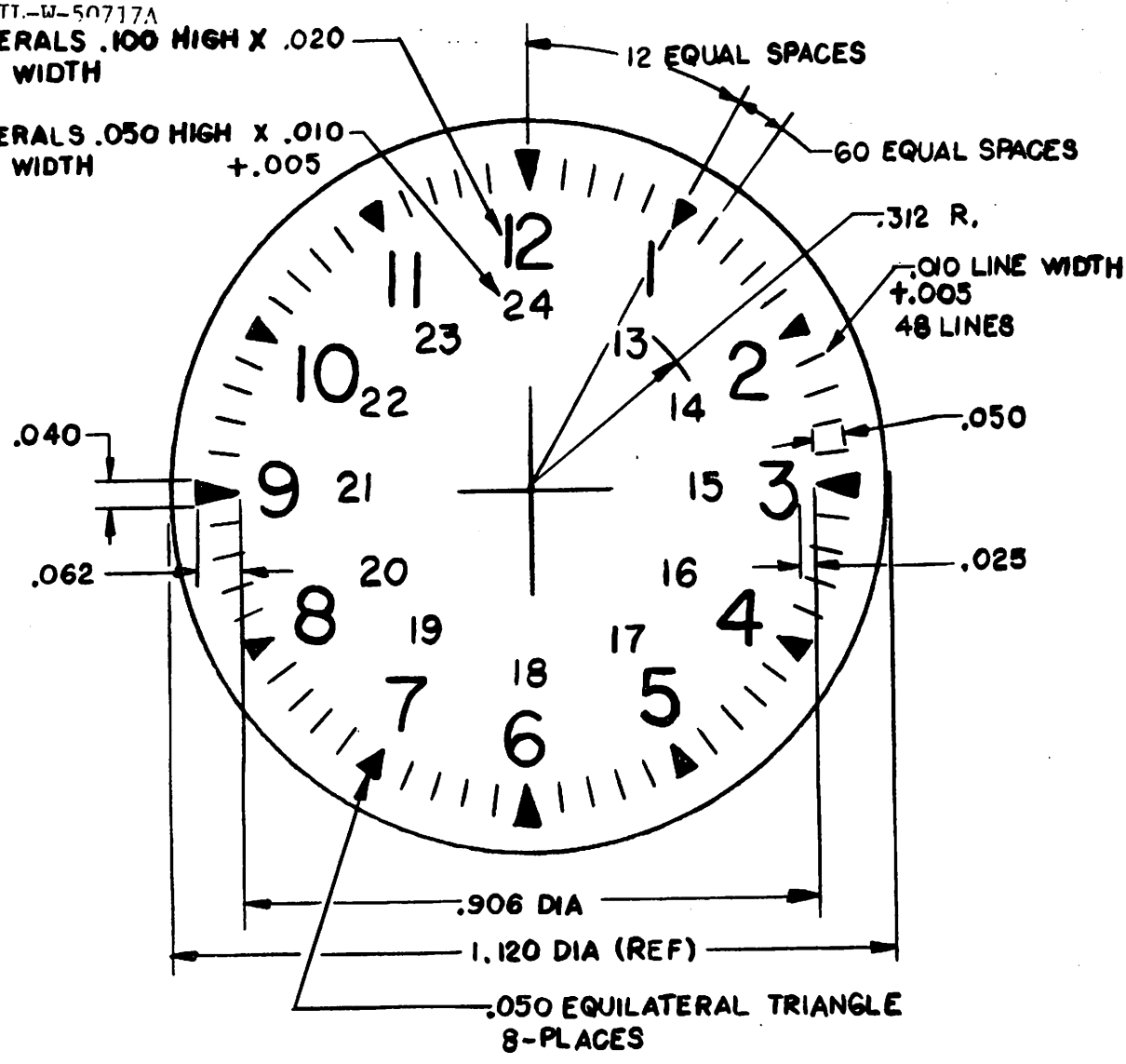
NOTES:

1. DIAL FACE BACKGROUND IN ACCORDANCE WITH FED-STD-595, COLOR BLACK NO.37038
2. ▨ = AREAS AND MINUTE GRADUATIONS TO BE IN ACCORDANCE WITH FED-STD-595, COLOR WHITE NO.37875
3. SHADED AREAS:
   CLASS A: LUMINOUS GREEN (SEE 3.3.2, 3.3.2.1 AND 3.3.2.2)
   CLASS B: FED-STD-595, COLOR WHITE NO. 37875

DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.

FIGURE 3

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TYPE II

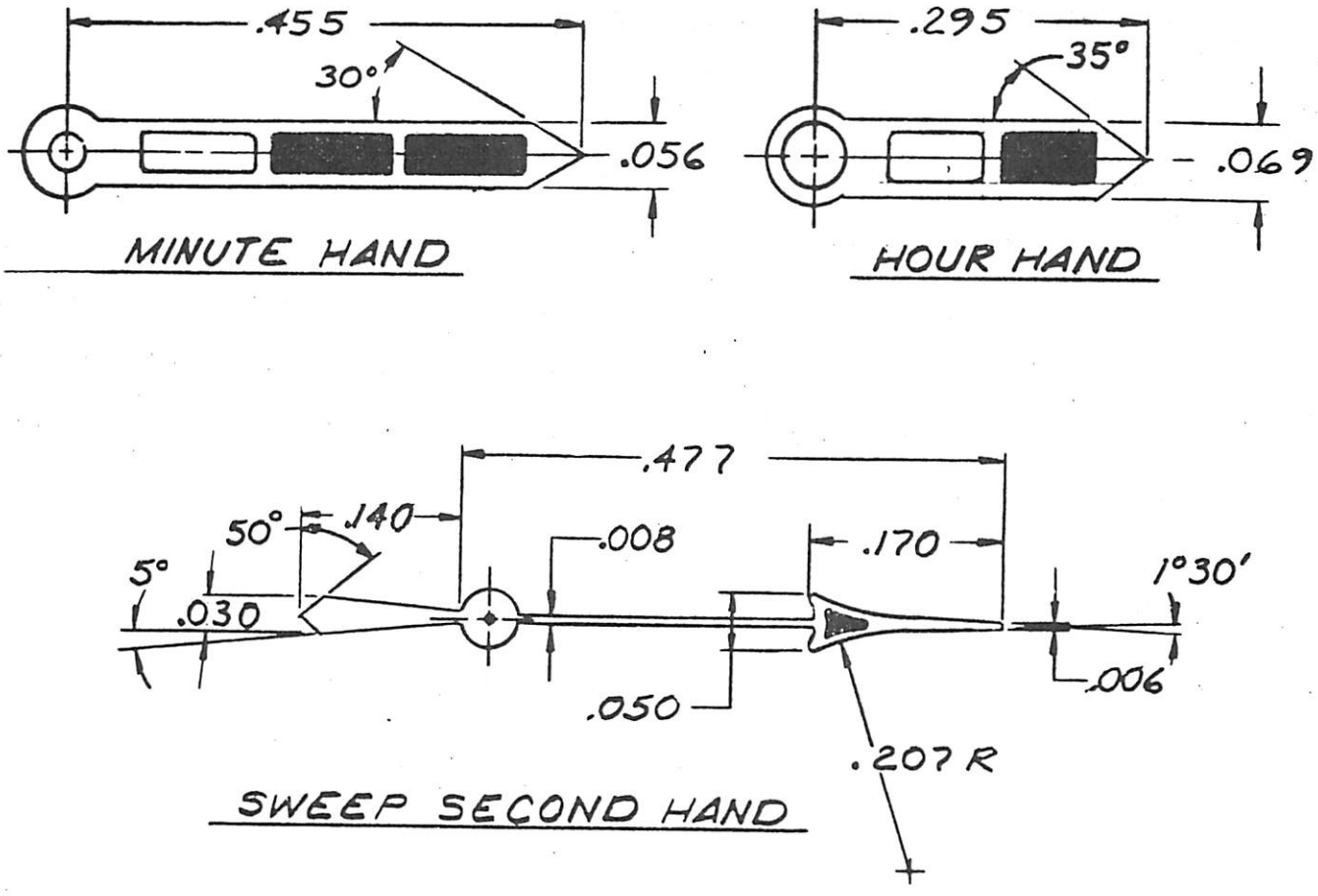
NOTES:

1. DIAL FACE BACKGROUND IN ACCORDANCE WITH FED-STD-595 COLOR BLACK NO.37038
2. NUMBERS AND GRADUATIONS IN ACCORDANCE WITH FED-STD-595, COLOR WHITE NO.37875
3. SHADED AREAS:
   CLASS A: LUMINOUS GREEN (SEE 3.3.2, 3.3.2.1, AND 3.3.2.2
   CLASS B: FED-STD-595, COLOR WHITE NO.37875

DIMENSIONS IN INCHES
UNLESS OTHERWISE
SPECIFIED.
TOLERANCES:
DECIMALS ± .010

FIGURE 4

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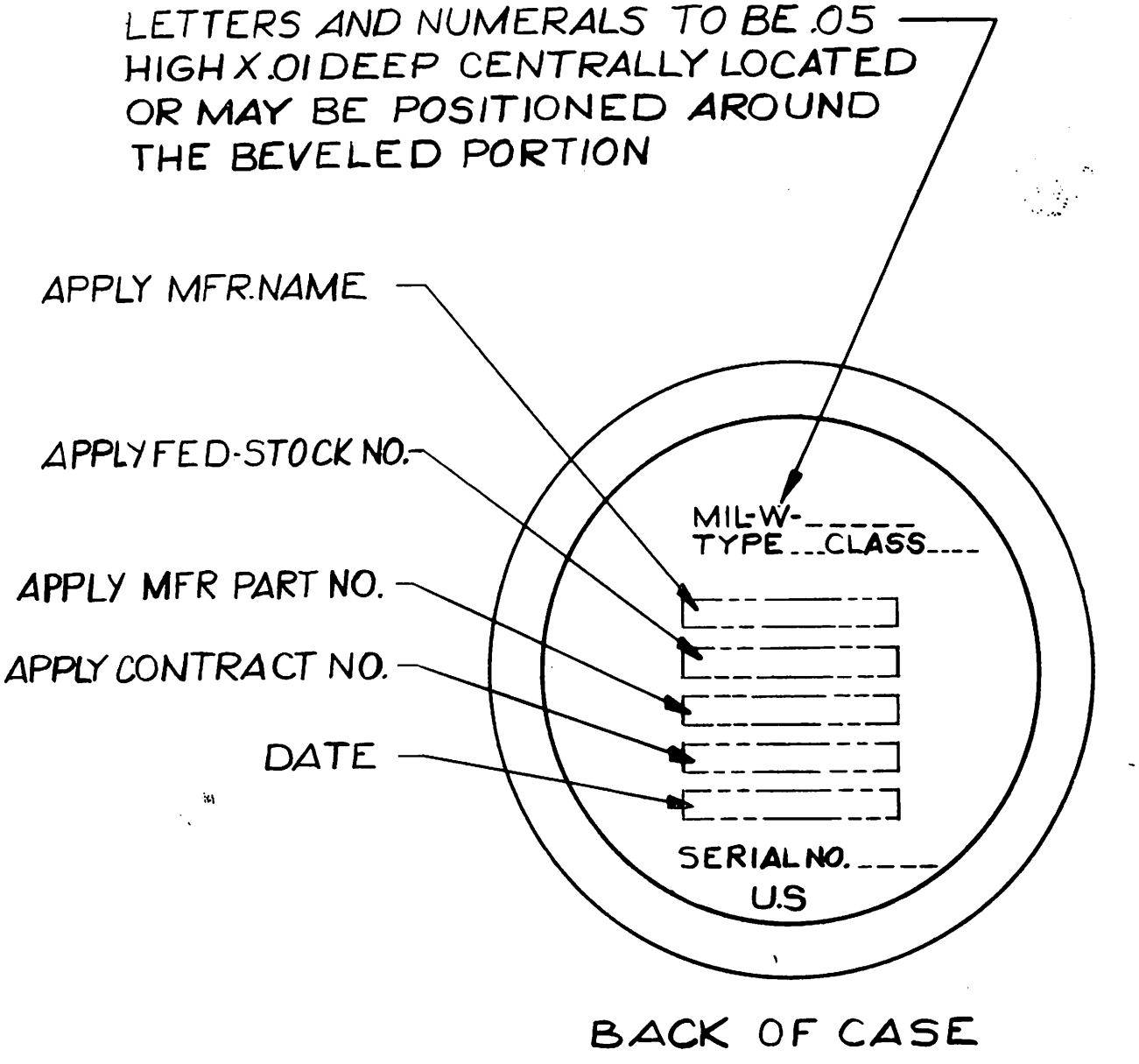
NOTES:

1. UNSHADED AREAS TO BE IN ACCORDANCE WITH FED-STD-595, COLOR WHITE NO.37875.
2. SHADED AREAS:
   CLASS A: LUMINOUS GREEN (SEE 3.3.2, 3.3.2.1 AND 3.3.22).
   CLASS B: FED-STD-595, COLOR WHITE NO.37875
3. THE HANDS FOR CLASS B WATCHES MAY BE SOLID HANDS, NOT SKELETON HANDS AS ILLUSTRATED.

DIMENSIONS IN INCHES
UNLESS OTHERWISE
SPECIFIED.
TOLERANCES:
DECIMALS ±.010

FIG. 5

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DIMENSIONS IN INCHES.
TOLERANCES ±.01.

FIG. 6

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